

Katherine Mao

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US Citizen

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Education

University of Pennsylvania – Philadelphia, PA
PhD in Mechanical Engineering
Advisors: M Ani Hsieh, Vijay Kumar

August 2021 - Present

University of Pennsylvania – Philadelphia, PA
Masters of Science in Engineering in Robotics

Dec 2023
GPA: 3.77

Purdue University - Honors College – West Lafayette, IN
Bachelor of Science in Mechanical Engineering
Minors: *Computer Science, Math*

May 2020
GPA: 3.77

Skills

Programming & Tools: Python, MATLAB, C/C++, CasADi, Pytorch

Technical: Trajectory/Motion Planning, Quadrotor Dynamics, Model-Based Control, Quaternions, Imitation Learning, Deep Learning

Mechanical Design/Manufacturing: SolidWorks, CATIA, Inventor, AutoCAD, Standard Shop Tools, Soldering

Languages: English, Chinese (Basic)

Publications

X. Yu, Y. Wu, **K. Mao**, Y. Tian, V. Kumar, T. Rosing, “DroneFL: Federated Learning for Multi-UAV Visual Target Tracking”, in 2026 IEEE International Conference on Robotics and Automation (ICRA). *In Submission*. [Link](#)

K. Mao, H.Yu, R. Zhang, I. Spasojevic, M. A. Hsieh, S. Gao, and V. Kumar, “Sequence Modeling for Time-Optimal Quadrotor Trajectory Optimization with Sampling-based Robustness Analysis”, in 2025 Conference on Robot Learning (CoRL). *Accepted*. [Link](#)

K. Mao, I. Spasojevic, M. Hopkins, M. A. Hsieh, and V. Kumar, “Collision-free time-optimal path parameterization for multi-robot teams”, *In Revision*

K. Mao, I. Spasojevic, M. A. Hsieh, and V. Kumar, “TOPPQuad: Dynamically-Feasible Time Optimal Path Parametrization for Quadrotors”, in 2024 IEEE International Conference on Intelligent Robots and Systems (IROS). *Accepted*. [Link](#)

K. Mao, J. Welde, M. A. Hsieh, and V. Kumar, “Trajectory planning for the bidirectional quadrotor as a differentially flat hybrid system” in 2023 IEEE International Conference on Robotics and Automation (ICRA), 2023, pp. 1242–1248. [Link](#)

N. Bajaj, N. Giampietro, **K. Mao**, et al. “Searching for Bed Bugs: The Design, Development, and Evaluation of an Oscillator-Based Trans-2-hexenal Sensor Array”. *Sensors & Actuators B*. vol. 129161. 2020

Posters and Presentations

Poster, Spotlight – Conference on Robot Learning (CoRL) 2025
“Sequence Modeling for Time-Optimal Quadrotor Trajectory Optimization with Sampling-based Robustness Analysis”

Poster, Lightning – Leveraging Implicit Methods for Aerial Autonomy Workshop @ Robotics: Science and System (RSS) 2025
“Sequence Modeling for Time-Optimal Quadrotor Trajectory Optimization with Sampling-based Robustness Analysis”
Invited Paper

Poster, Lightning – The Institute for Learning-enabled Optimization at Scale (TILOS) Industry Day 2025
“Sequence Modeling for Time-Optimal Quadrotor Trajectory Optimization with Sampling-based Robustness Analysis”

Oral – IEEE International Conference on Intelligent Robots and Systems (IROS) 2024
“TOPPQuad: Dynamically-Feasible Time Optimal Path Parametrization for Quadrotors”
Accepted as Oral Presentation

Poster, Lightning – The Institute for Learning-enabled Optimization at Scale (TILOS) Industry Day 2024

“TOPPQuad: Dynamically-Feasible Time Optimal Path Parametrization for Quadrotors”

Poster, Lightning – The Institute for Learning-enabled Optimization at Scale (TILOS) Industry Day 2023
“Trajectory planning for the bidirectional quadrotor as a differentially flat hybrid system”

Poster – IEEE International Conference on Robotics and Automation (ICRA) 2023
“Trajectory planning for the bidirectional quadrotor as a differentially flat hybrid system”

Work and Research Experience

Bolt Medical, Systems Team | *Systems Design Engineer* July 2020 – August 2021

- **Project details under NDA**
- Wrote OpenCV code to determine moving object location with micrometer accuracy
- Designed Python GUI for combined automation and control of components within prototype system
- Characterized and Isolated critical sensor feedback to determine breakage within prototype system
- Compiled and assembled wiring layout for the prototype system

Purdue University, Rhoads Group | *Research Assistant* January 2018 – May 2020

- Manufactured PCBs for lead detection with microelectromechanical system (MEMs) sensors
- Designed controller and GUI for instrumentation and data analysis of lead sensors in MATLAB
- Wrote jogging controller and GUI for laser printer within Arduino platform
- Designed testing chamber for chemical bed bug detection method
- Wrote FPGA frequency counter and UART transmission in Verilog and Quartus Prime
- Wrote data collection and visualizer tool for a (field-programmable gate array) FPGA frequency counter in Python

HP Inc, Writing Systems | *Product Design Intern* May 2019 – Aug 2019

- Designed diagnostics tool to analyze sub-millimeter deformation along printbar in global industrial printers
- Integrated Arduino microcontroller, IR sensors, Inertial Measurement Unit into 3D printed enclosure
- Developed tool code for data logging, calibration, and analysis in Arduino and Python
- Improved runtime and data acquisition of printbar deformation from existing techniques by an order of magnitude

Johns Hopkins University, Whitcomb Group | *Research Assistant* May 2017 - Aug 2017, May 2018 – Aug 2018

- Research Experience for Undergraduates in Computational Sensing and Medical Robotics (CSMR REU)
- Collaborated with graduate students on JHU ROV II (Remotely Operated underwater Vehicle) testbed design
- Designed waterproof connector adapter for titanium housing rated to 500m depth in SolidWorks
- Manufactured hardware for a 3' watertight enclosure with lathe, mill, and other shop tools
- Submitted project report to NSF and presented final research to all principal investigators

Purdue University, IEEE ROV Team | *Logic and Structures Project Lead* September 2016 – May 2019

- Designed the 2017 and 2018 competition ROV frame, thruster, and secondary camera mounts in SolidWorks
- Recipient of the Flying Fish Award for Ingenuity and Creativity in Design at the 2017 MATE International Competition
- Contributed to paper published in IEEE Computer Society Magazine (Vol 50, Issue 9, 2017)

Academic Honors

3rd Place – Ecomake Hackathon, *Purdue University* 2018

Kaiser Aluminum Mechanical Engineering Scholarship, *Purdue University* 2018

Deans List Recipient, *Purdue University* 2016, 2017, 2018, 2019, 2020

Flying Fish Award for Ingenuity and Creativity in Design, *2017 MATE International Competition* 2017

Presidential Scholarship Recipient, *Purdue University* 2016

National Merit Finalist, *NSMC* 2016

Affiliations: FIRST, IEEE, National Center for Women in Technology (NCWIT), SWE

Interests: Sketching/Digital Art, Fencing, Origami/Kusudama/Papercraft, Language Learning, Bookbinding